GCSE BIOLOGY

Higher Tier – Paper 1 style

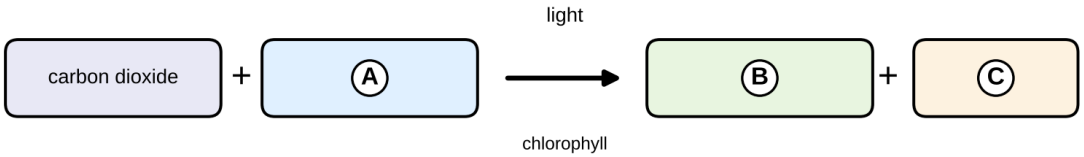
Topic 4.4: Bioenergetics

Practice questions in the style of AQA GCSE Biology (8461). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 165 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **159**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 159

0 1 Photosynthesis happens in the leaves of plants. The diagram shows the word equation for photosynthesis with some parts missing.



Symbol equation: $6CO_2$ + 6___ → ___ + 6___

0 1 . 1 Name the substances **A**, **B** and **C**. **[3 marks]**

0 1 . 2 Complete the symbol equation for photosynthesis shown in the diagram. **[3 marks]**

0 1 . 3 Photosynthesis is an **endothermic** reaction. What does **endothermic** mean? **[1 mark]**

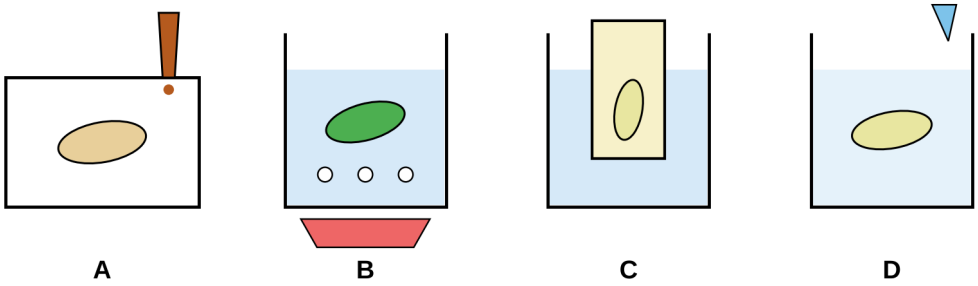
0 1 . 4 State the source of the energy for photosynthesis. **[1 mark]**

0 1 . 5 Name the part of a plant cell where photosynthesis happens and the green substance that absorbs the light. **[2 marks]**

0 1 . 6 Name **two** ways in which a leaf is adapted to obtain carbon dioxide and water for photosynthesis. **[2 marks]**

0 2 A student tests a leaf for the presence of starch. The diagram shows four stages, **A** to **D**, of the method. They are **not** in the correct order.

0 2 . 1 Put stages **A** to **D** in the correct order. **[2 marks]**



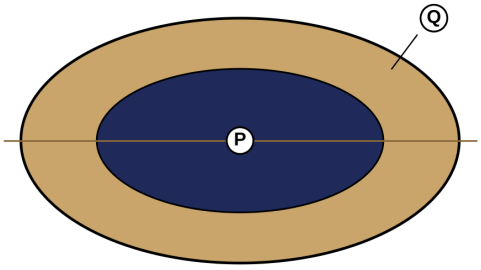
0 2 . 2 Explain why the leaf is placed in boiling water at the start. **[2 marks]**

0 2 . 3 The leaf is then heated in ethanol. Explain why. **[2 marks]**

0 2 . 4 Ethanol is flammable. Describe how the student should heat the ethanol safely. **[2 marks]**

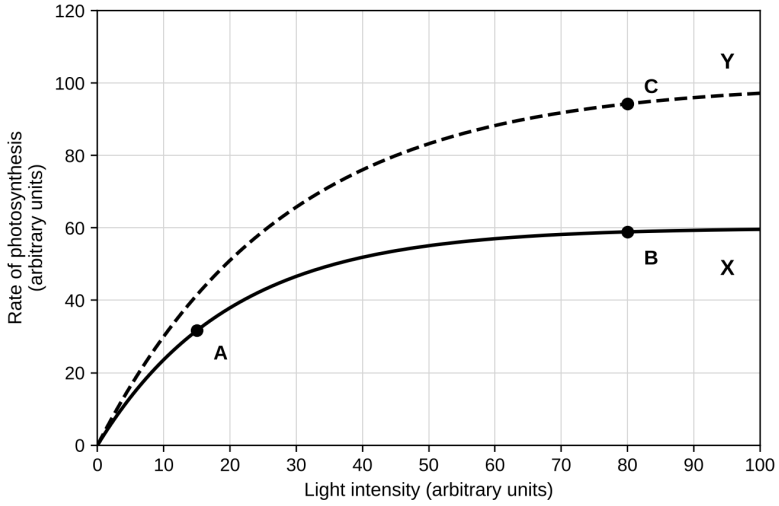
0 2 . 5 State the colour of the leaf when iodine solution is added to a leaf containing starch. **[1 mark]**

0 2 . 6 A leaf that is green in the middle and white at the edges was tested for starch. The diagram shows the leaf after iodine was added. Explain the results at **P** and **Q**. **[3 marks]**



0 2 . 7 Explain why a plant that was kept in the dark for 48 hours before the test showed no starch in its leaves. **[2 marks]**

0 3 A scientist investigates the effect of light intensity on the rate of photosynthesis in pondweed at 25 °C. She uses two different concentrations of carbon dioxide: 0.04 % (**X**) and 0.12 % (**Y**). The graph shows her results.



0 3 . 1 What is the independent variable in this investigation? **[1 mark]**

0 3 . 2 Use curve **X** to find the rate of photosynthesis at a light intensity of 20 units. **[1 mark]**

0 3 . 3 What is the limiting factor at point **A**? Give a reason. **[2 marks]**

0 3 . 4 What is the limiting factor at point **B**? Give a reason. **[2 marks]**

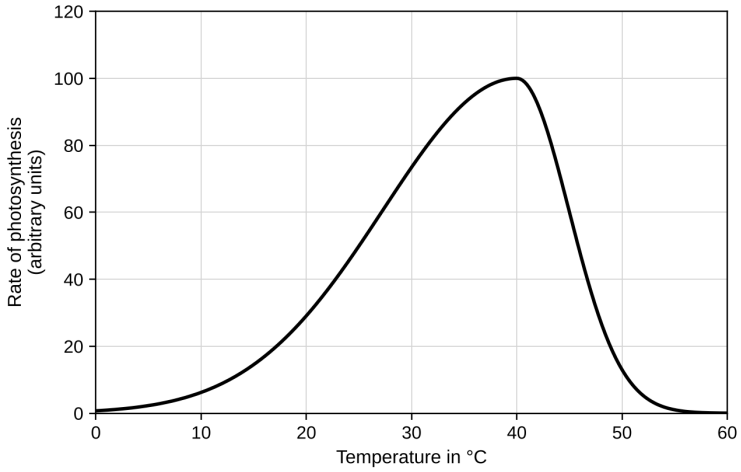
0 3 . 5 Calculate the percentage increase in the rate of photosynthesis from **B** to **C**. **[3 marks]**

Answer = _____ %

0 3 . 6 The scientist repeats the investigation at 15 °C. Predict what happens to the plateau of curve **X**. Give a reason. **[2 marks]**

0 3 . 7 Name **two** other factors that can limit the rate of photosynthesis. **[2 marks]**

0 4 The graph shows how the rate of photosynthesis changes with temperature in a plant.



0 4 . 1 Describe the change in the rate of photosynthesis between 10 °C and 40 °C. **[2 marks]**

0 4 . 2 Use the graph to find the optimum temperature for photosynthesis. **[1 mark]**

0 4 . 3 Explain why the rate increases as the temperature rises from 10 °C to 40 °C. **[2 marks]**

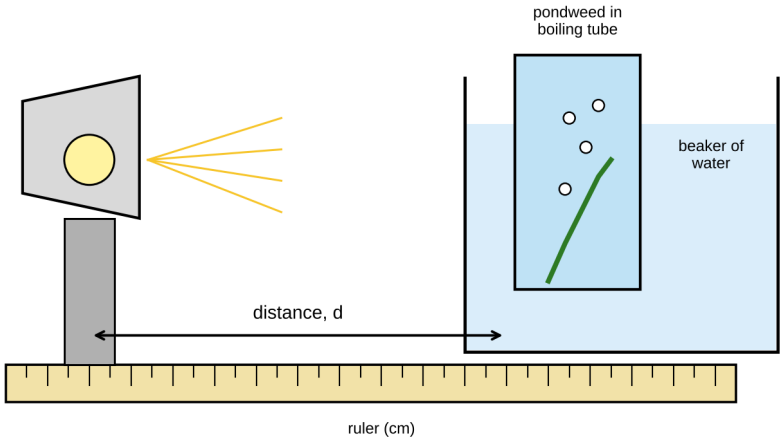
0 4 . 4 Explain why the rate falls above 40 °C. **[3 marks]**

0 4 . 5 How many times greater is the rate at 30 °C than the rate at 20 °C? Use values from the graph. **[2 marks]**

Answer = _____ times

0 4 . 6 Suggest why the rate of photosynthesis does not increase beyond a certain temperature, even when light and carbon dioxide are plentiful. **[1 mark]**

0 5 A student investigates how the distance of a lamp from a plant affects the rate of photosynthesis. She places a lamp at different distances from pondweed. She counts the number of bubbles of oxygen given off in one minute. The table shows her results.



0 5 . 1 Complete the table by calculating the number of bubbles per minute at 30 cm if the rate at 10 cm is 64 and the rate follows an inverse square law. (Light intensity ∝ 1 ÷ distance², so rate at 30 cm = 64 × (10 ÷ 30)².) **[2 marks]**

Distance of lamp in cm	**10**	**20**	**30**	**40**	**50**
Bubbles per minute	64	16		4	2.6

Answer = _____ bubbles

0 5 . 2 The student counts bubbles of gas. Suggest **one** reason why this does not give an accurate measure of the rate of photosynthesis. **[1 mark]**

0 5 . 3 Explain why the rate of photosynthesis decreases as the lamp is moved further away. **[2 marks]**

0 5 . 4 If the distance doubles from 10 cm to 20 cm, the light intensity falls to 1/4. Calculate the light intensity at 40 cm if it is 400 units at 10 cm. **[3 marks]**

Answer = _____ units

0 5 . 5 Name **two** variables the student should keep constant. **[2 marks]**

0 5 . 6 The lamp makes the water warmer as it gets closer. Suggest how the student could stop this affecting the results. **[1 mark]**

0 6 A farmer grows tomatoes in glasshouses. He can control the conditions in the glasshouse. The table shows the cost and the yield of tomatoes for three different glasshouse conditions each month. Tomatoes are sold for £1.20 per kilogram.

0 6 . 1 Complete the table to show the profit for each condition. **[3 marks]**
Profit = income − cost

Condition	Cost per month in £	Yield in kg	Profit in £
A: heating only	800	1200	
B: heating, extra light and carbon dioxide	1500	2100	
C: heating and extra light	1100	1500	

Answer = _____ £

0 6 . 2 Which condition gives the highest profit? Give **one** reason why the farmer might still choose another condition. **[2 marks]**

0 6 . 3 Explain why adding extra light increases the yield. **[2 marks]**

0 6 . 4 The farmer burns a fuel in the glasshouse to produce carbon dioxide and heat. Explain how this increases the yield. **[3 marks]**

0 6 . 5 Suggest why painting the inside of a glasshouse wall white can increase yield. **[1 mark]**

0 7 Plants make glucose by photosynthesis. Glucose is used in several ways.

0 7 . 1 Complete the table to show how plants use glucose. **[4 marks]**

Use of glucose	**What it is used for**
	releases energy for the plant
converted to insoluble starch	stored in roots / leaves / seeds
converted to cellulose	
	cell walls

0 7 . 2 Explain why plants store glucose as starch and not as glucose. **[2 marks]**

0 7 . 3 Plants use glucose and nitrate ions to make amino acids. Name the substances that amino acids are used to make. **[1 mark]**

0 7 . 4 Name **two** other substances plants make from glucose that are used as energy stores in seeds. **[2 marks]**

0 7 . 5 Explain why a potato plant stores glucose as starch in its tubers. **[2 marks]**

0 8 Respiration happens in all living cells.

0 8 . 1 What is meant by **respiration**? **[2 marks]**

0 8 . 2 Respiration is an **exothermic** reaction. What is meant by **exothermic**? **[1 mark]**

0 8 . 3 Write the word equation for aerobic respiration. **[2 marks]**

0 8 . 4 Write the balanced symbol equation for aerobic respiration. **[2 marks]**

0 8 . 5 Name the part of the cell where aerobic respiration happens. **[1 mark]**

0 8 . 6 Describe **three** ways in which organisms use the energy released by respiration. **[3 marks]**

0 8 . 7 Plants respire all the time but only photosynthesise in the light. Explain why. **[2 marks]**

0 9 Respiration can be aerobic or anaerobic.

0 9 . 1 What is meant by **anaerobic respiration**? **[1 mark]**

0 9 . 2 Complete the table to compare aerobic and anaerobic respiration in humans. **[4 marks]**

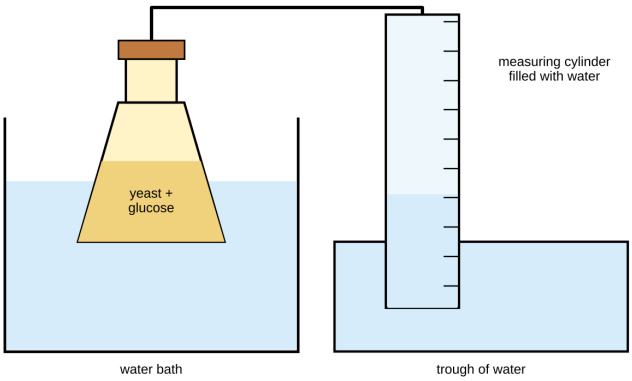
	Aerobic respiration	**Anaerobic respiration in muscles**
Word equation		
Amount of energy released per glucose molecule		

0 9 . 3 Write the word equation for anaerobic respiration in yeast. **[2 marks]**

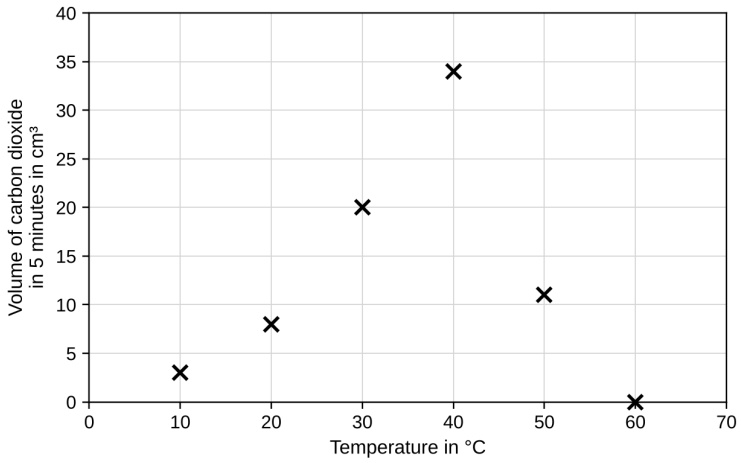
0 9 . 4 What is the name of the process by which yeast respires anaerobically? **[2 marks]**
Give **one** use of this process.

0 9 . 5 Explain why bacteria in waterlogged soil respire anaerobically. **[1 mark]**

1 0 A student investigates the effect of temperature on the rate of anaerobic respiration in yeast. She mixes yeast with glucose solution and places the flask in a water bath. She measures the volume of carbon dioxide produced in 5 minutes. She repeats this at six temperatures. The apparatus and her results are shown.



1 0 . 1 Draw a curve of best fit on the graph through the points. **[2 marks]**



1 0 . 2 Calculate the rate of carbon dioxide production at 40 °C in cm³ per minute. **[2 marks]**

Answer = _____ cm³/min

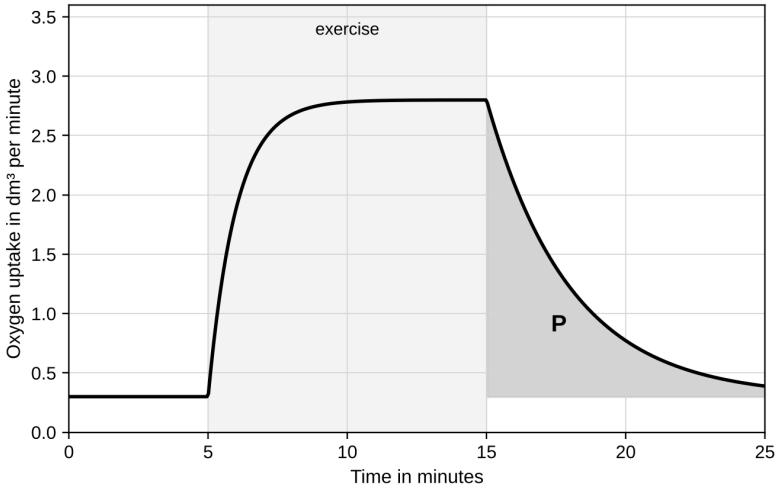
1 0 . 3 Explain why the volume of carbon dioxide produced is lower at 50 °C than at 40 °C. **[3 marks]**

1 0 . 4 Name **two** variables the student must control. **[2 marks]**

1 0 . 5 Suggest why the student must let the yeast and glucose solution reach the water-bath temperature before measuring. **[1 mark]**

1 0 . 6 Suggest an improvement to the method that would make the results more reliable. **[1 mark]**

1 1 A person runs on a treadmill for 10 minutes. The graph shows the oxygen uptake of the person before, during and after the exercise.



1 1 . 1 State the person's oxygen uptake at 10 minutes. **[1 mark]**

1 1 . 2 Describe **two** changes in the body during exercise that increase the supply of oxygen to the muscles. **[2 marks]**

1 1 . 3 The person's oxygen uptake is still above resting level 10 minutes after exercise has stopped. Explain why. **[3 marks]**

1 1 . 4 What does area **P** represent? **[1 mark]**

1 1 . 5 Explain why muscles may become tired and stop contracting efficiently after long periods of vigorous exercise. **[2 marks]**

1 1 . 6 Describe what happens to the lactic acid after exercise. **[2 marks]**

1 1 . 7 Explain why a trained athlete recovers more quickly than an untrained person. **[2 marks]**

1 2 Metabolism is the sum of all the chemical reactions in a cell or the body.

1 2 . 1 Give **two** examples of metabolic reactions in which small molecules are joined to make larger molecules. **[2 marks]**

1 2 . 2 Name the products formed when lipids are broken down. **[2 marks]**

1 2 . 3 What is the name of the process by which excess amino acids are broken down? **[1 mark]**

12.4 Name the waste substance formed when proteins are broken down and state where it is made. **[2 marks]**

12.5 Glucose is converted to glycogen in the liver and muscles. Explain why this is useful during exercise. **[2 marks]**

12.6 Explain why metabolic reactions need enzymes. **[2 marks]**

13 Stretch questions (grade 7–9).

13.1 *Explain how light intensity, carbon dioxide concentration and temperature each affect the rate of photosynthesis. Use the idea of limiting factors in your answer. **[6 marks]**

1 3 . 2 *A farmer wants to increase the profit from a glasshouse of tomato plants. Evaluate the use of additional heating, lighting and carbon dioxide. **[6 marks]**

1 3 . 3 *Compare aerobic and anaerobic respiration in humans. Explain why the body uses anaerobic respiration during vigorous exercise and why an oxygen debt builds up. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4.4: Bioenergetics

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	A: water ; B: glucose ; C: oxygen ;		3	AO1 4.4.1.1
01.2	H_2O ; $C_6H_{12}O_6$; O_2 ;	$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. Balancing numbers must be correct (6 H_2O and 6 O_2).	3	AO1 4.4.1.1
01.3	energy is transferred from the surroundings (to the chloroplasts) ;	Allow: takes in energy / light.	1	AO1 4.4.1.1
01.4	light (from the Sun)		1	AO1 4.4.1.1
01.5	chloroplast ; chlorophyll ;		2	AO1 4.4.1.1
01.6	stomata let carbon dioxide diffuse in ; water is brought to the leaf by the xylem (from the roots) ;	Allow: large surface area / air spaces in the spongy layer.	2	AO1 4.4.1.1
02.1	B, C, D, A ; (three in the correct order for 1 mark) ;	B: boiling water kills the leaf; C: hot ethanol removes chlorophyll; D: rinse softens the leaf; A: iodine.	2	AO2 4.4.1.2 / RP
02.2	to kill the leaf / stop chemical reactions ; and soften it so the iodine can enter the cells ;		2	AO2 4.4.1.2 / RP
02.3	ethanol removes the chlorophyll / decolourises the leaf ; so the colour change with iodine can be seen ;		2	AO2 4.4.1.2 / RP
02.4	use a hot water bath / electric heater (not a naked flame) ; turn off Bunsen burners nearby / wear eye protection ;		2	AO1 4.4.1.2 / RP
02.5	blue-black		1	AO1 4.4.1.2 / RP
02.6	P (green): chlorophyll present so photosynthesis happens and starch is stored ; Q (white): no chlorophyll so no photosynthesis / no starch ; iodine stays brown / orange in Q ;		3	AO3 4.4.1.2 / RP
02.7	no light so no photosynthesis ; starch used up / respired / moved away ;		2	AO3 4.4.1.2 / RP
03.1	light intensity (allow: concentration of carbon dioxide)		1	AO3 WS 2.3
03.2	36–40 (arbitrary units)		1	AO3 WS 3.1
03.3	light intensity ; increasing the light intensity increases the rate / the curve is still rising ;		2	AO3 4.4.1.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	carbon dioxide (concentration) ; increasing light intensity does not increase the rate / the rate is higher at C with more CO_2 ;	Allow temperature as a possible second factor if a sensible reason is given.	2	AO3 4.4.1.3
03.5	C = 94 and B = 59 (allow ±2) ; (94 − 59) ÷ 59 × 100 ; 59 % (allow 55–63) ;		3	AO2 MS 1c
03.6	the plateau is lower ; enzymes work more slowly at lower temperature ;		2	AO3 4.4.1.3
03.7	temperature ; amount of chlorophyll ;		2	AO1 4.4.1.3
04.1	rate increases ; from about 6 to 100 arbitrary units / quote data ;		2	AO3 4.4.1.3
04.2	40 (°C)		1	AO3 WS 3.1
04.3	particles / molecules have more kinetic energy ; so more collisions between enzymes and substrates (more enzyme-controlled reactions) ;		2	AO2 4.4.1.3
04.4	enzymes (for photosynthesis) are denatured ; the shape of the active site changes ; so substrates no longer fit / reaction slows ;		3	AO2 4.4.1.3
04.5	about 73 and 29 ; 2.5 (allow 2.2–2.8) ;		2	AO2 MS 1a
04.6	enzymes denature / another factor becomes limiting ;		1	AO3 4.4.1.3
05.1	64 × (10/30)² ; 7.1 (allow 7) ;	Triple only (inverse square law). Allow 7 bubbles.	2	AO2 4.4.1.4
05.2	bubble size varies / gas dissolves in water / some bubbles are not counted ;	Allow: collecting the gas in a gas syringe would be more accurate.	1	AO3 4.4.1.3 / RP
05.3	light intensity decreases as distance increases ; so less energy absorbed by chlorophyll / light is the limiting factor ;		2	AO2 4.4.1.3 / RP
05.4	(10 ÷ 40)² = 1/16 ; 400 ÷ 16 ; 25 units ;		3	AO2 4.4.1.4
05.5	temperature of the water / use a heat shield / water bath ; same pondweed / same size piece / carbon dioxide concentration ;	Allow: lamp (type and power), time.	2	AO3 4.4.1.3 / RP
05.6	place a beaker of water / heat shield between the lamp and pondweed / use LED lamp ;		1	AO3 4.4.1.3 / RP
06.1	A: 1200 × 1.20 = £1440; 1440 − 800 = £640 ; B: 2100 × 1.20 = £2520; 2520 − 1500 = £1020 ; C: 1500 × 1.20 = £1800; 1800 − 1100 = £700 ;		3	AO2 4.4.1.5 / MS 1a
06.2	B ; (e.g.) higher initial cost / may not have enough money to pay for it / environmental cost ;		2	AO3 4.4.1.5

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3	light is a limiting factor / more light increases the rate of photosynthesis ; so more glucose is made for growth ;		2	AO2 4.4.1.5
06.4	carbon dioxide is a limiting factor so adding more increases photosynthesis ; heat keeps the temperature close to the optimum for enzymes ; faster growth means more tomatoes ;		3	AO2 4.4.1.5
06.5	reflects light back onto the plants / more light for photosynthesis ;		1	AO3 4.4.1.5
07.1	respiration ; energy / transfers energy for growth ; starch ; cellulose ;	Allow answers in any sensible form: respiration (releases energy); starch (store); cellulose (cell walls); fats / oils (energy store in seeds); amino acids with nitrate ions (proteins).	4	AO1 4.4.1.6
07.2	glucose is soluble ; so it would affect osmosis / starch is insoluble so does not alter the water concentration ;		2	AO2 4.4.1.6
07.3	proteins		1	AO1 4.4.1.6
07.4	fats ; oils ;		2	AO1 4.4.1.6
07.5	so there is a store of energy for the next growing season ; starch is insoluble so does not cause water to enter by osmosis ;		2	AO3 4.4.1.6
08.1	a chemical reaction that happens in cells ; that transfers / releases energy from glucose / nutrient molecules ;		2	AO1 4.4.2.1
08.2	transfers energy to the surroundings (as heat)		1	AO1 4.4.2.1
08.3	glucose + oxygen ; → carbon dioxide + water ;		2	AO1 4.4.2.1
08.4	$C_6H_{12}O_6 + 6O_2$; → $6CO_2 + 6H_2O$;		2	AO1 4.4.2.1
08.5	mitochondria / mitochondrion		1	AO1 4.4.2.1
08.6	muscle contraction / movement ; keeping warm (maintaining body temperature in mammals and birds) ; building larger molecules (e.g. proteins / amino acids, DNA) ;	Allow: active transport. Any three.	3	AO1 4.4.2.1
08.7	respiration releases energy needed for all life processes ; photosynthesis needs light energy ;		2	AO2 4.4.2.1
09.1	respiration without oxygen		1	AO1 4.4.2.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
09.2	glucose + oxygen → carbon dioxide + water ; much more energy / full oxidation of glucose ; glucose → lactic acid ; less energy released ;	Two marks per row, one for each correct cell.	4	AO1 4.4.2.1 / 4.4.2.2
09.3	glucose → ethanol ; + carbon dioxide ;		2	AO1 4.4.2.1
09.4	fermentation ; making bread (CO_2 makes dough rise) / making alcoholic drinks (ethanol) ;		2	AO1 4.4.2.1
09.5	there is little / no oxygen in waterlogged soil (air spaces filled with water) ;		1	AO3 4.4.2.1
10.1	smooth curve ; passes close to / through most points (curve rises then falls) ;	Do not accept dot-to-dot lines.	2	AO2 WS 3.5
10.2	34 ÷ 5 ; 6.8 cm^3/min ;		2	AO2 MS 1a
10.3	enzymes in the yeast are denatured ; shape of the active site is changed ; so respiration reaction is slower / glucose cannot bind ;		3	AO2 4.4.2.1 / RP
10.4	volume / concentration of glucose solution ; mass / concentration of yeast ;	Allow: time; pH.	2	AO3 WS 2.3
10.5	so the temperature of the mixture is the correct temperature during the measurement ;		1	AO3 RP
10.6	repeat each temperature (3 times) and calculate a mean ;		1	AO3 WS 3.4
11.1	2.8 dm^3 per minute	Allow 2.7–2.9.	1	AO3 WS 3.1
11.2	heart rate increases ; breathing rate / depth of breathing increases ;	Allow: more blood flows to muscles.	2	AO1 4.4.2.2
11.3	during exercise some anaerobic respiration happened ; lactic acid built up (an oxygen debt) ; extra oxygen is needed after exercise to break down lactic acid / convert it to glucose in the liver ;		3	AO2 4.4.2.2
11.4	the oxygen debt / excess post-exercise oxygen consumption ;		1	AO3 4.4.2.2
11.5	lactic acid builds up (muscle fatigue) ; because blood does not supply oxygen quickly enough for aerobic respiration ;		2	AO2 4.4.2.2
11.6	it is transported to the liver in the blood ; converted back into glucose (or broken down with oxygen) ;		2	AO1 4.4.2.2
11.7	their heart / lungs deliver oxygen faster (higher oxygen uptake) ; so the oxygen debt is repaid more quickly ;		2	AO3 4.4.2.2
12.1	glucose → starch / glycogen / cellulose ; glycerol + 3 fatty acids → a lipid ;	Allow: amino acids → proteins; nitrate + glucose → amino acids.	2	AO1 4.4.2.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
12.2	fatty acids ; glycerol ;		2	AO1 4.4.2.3
12.3	deamination (accept breakdown in the liver)		1	AO1 4.4.2.3
12.4	urea ; liver ;		2	AO1 4.4.2.3
12.5	glycogen is a store of glucose ; it is broken down to glucose for respiration when energy is needed ;		2	AO2 4.4.2.3
12.6	enzymes speed up reactions / are biological catalysts ; at body temperature reactions would be too slow ;		2	AO2 4.4.2.3
13.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • limiting factor = the factor in shortest supply that stops the rate rising • light intensity: more light increases the rate until another factor limits it; graph levels off • carbon dioxide: more CO2 raises the rate up to a maximum; it is needed as a reactant • temperature: rate rises to an optimum (about 40 °C) as enzyme activity increases • above the optimum, enzymes denature and the rate falls rapidly • factors interact: e.g. increasing light only helps if CO2 and temperature are not limiting		6	AO1 4.4.1.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
13.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • each factor can be a limiting factor so increasing them can increase the rate of photosynthesis and yield • heating keeps the temperature near the optimum for enzymes; useful in cold weather • extra light allows photosynthesis at night / in winter, a longer growing season • carbon dioxide enrichment increases the rate when CO2 is limiting • costs: fuel / electricity; initial equipment cost; a balance between cost and extra income is needed • conclusion based on data: profit only rises if extra yield × price is greater than the extra cost		6	AO3 4.4.1.5
13.3	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • aerobic: glucose + oxygen → carbon dioxide + water; much more energy released; occurs in mitochondria • anaerobic: glucose → lactic acid; less energy released per glucose; no oxygen needed • during vigorous exercise muscles need energy faster than oxygen can be supplied • anaerobic respiration releases energy quickly even without enough oxygen • lactic acid builds up and can cause muscle fatigue; blood carries it to the liver • extra oxygen is needed after exercise to break down lactic acid: this is the oxygen debt; heart / breathing rates stay high		6	AO2 4.4.2.1 / 4.4.2.2

Where marks are lost in Topic 4.4

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Photosynthesis:** $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (endothermic, needs light and chlorophyll). Learn it and check balancing numbers.
- ☐ **Limiting factors:** a rising curve means the factor on the x-axis is limiting; a flat curve means a **different** factor (CO_2, temperature) is limiting. Name the factor and give a reason from the graph.
- ☐ **Temperature graph:** rises because of more kinetic energy and enzyme collisions; falls above the optimum because enzymes **denature**.
- ☐ **Inverse square law:** intensity $\propto 1 \div d^2$. Doubling the distance gives ¼ of the intensity; tripling gives 1/9.
- ☐ **Starch test:** boiling water (kill) → hot ethanol (remove chlorophyll, use a water bath) → rinse → iodine (blue-black = starch).
- ☐ **Respiration:** glucose + oxygen → carbon dioxide + water (mitochondria). It **releases** energy — it does not 'make' energy.
- ☐ **Anaerobic:** muscles make lactic acid; yeast makes ethanol + carbon dioxide (fermentation). Less energy is released than in aerobic respiration.
- ☐ **Oxygen debt:** extra oxygen needed after exercise to break down lactic acid; heart and breathing rate stay high until it is repaid.
- ☐ **Metabolism:** learn three build-up reactions (starch / cellulose, lipids, proteins) and the breakdown product urea.
- ☐ **Economics (glasshouse):** profit = income − cost. Always show both calculations and compare.